Madhuben & Bhanubhai Patel Institute Of Technology

The Charutar Vidya Mandal(CVM) University

New V.V. Nagar

COMPUTER ENGINEERING DEPARTMENT

A.Y 2025-2026 EVEN TERM

Mini Project Report

SMART HELMET FOR MINORS - MineGUARD

Under

MINOR DEGREE PROGRAM
Industry 4.0 and IIoT(202170601)

Submitted by

Krish Mahendrabhai Kushvaha (12302040701086)

Pawan Jigneshkumar Sharma (12302040701130)

Poojan Daxeshbhai Mistry (12302080701097)

Guided by: Prof. Tejal Tandel

ABSTRACT

The mining industry serves as a critical foundation for global industrial development; however, it remains one of the most hazardous occupations worldwide. Workers are frequently exposed to life-threatening conditions such as toxic gas emissions, sudden structural collapses, and unmonitored environmental changes, leading to numerous fatalities each year. Conventional safety systems are largely reactive and suffer from delayed response times, making them insufficient for preventing localized underground disasters.

To overcome these limitations, this project presents the design, development, and evaluation of an IoT-based Smart Safety Helmet specifically tailored for hazardous underground mining environments. The proposed system integrates an ESP8266 NodeMCU microcontroller with environmental sensors, including the MQ-4 gas sensor for methane detection and the DHT11 sensor for real-time temperature and humidity monitoring.

A key innovation of this system is the implementation of SX1278 LoRa (Long Range) point-to-point communication, which effectively addresses the limitations of traditional wireless technologies such as Wi-Fi and Bluetooth that fail to operate reliably through dense underground rock structures.

The Smart Helmet continuously monitors environmental parameters and compares them against predefined safety thresholds. Upon detecting hazardous methane concentrations (above 1000 ppm) or extreme temperature conditions (above 45°C), the system triggers an immediate dual-response mechanism. This includes activating an onboard acoustic alarm (85 dB) to alert the miner and transmitting an emergency signal via LoRa to a remote base station located up to 3 kilometers away.

Additionally, the system is integrated with a cloud-based monitoring platform (ThingSpeak), enabling real-time data visualization and remote supervision. Experimental results demonstrate a rapid alert response time of under 2 seconds, reliable long-range communication, and efficient power consumption, allowing continuous operation for 48 to 72 hours on a 3000mAh battery.

With an estimated cost of approximately ₹3,000 per unit, the proposed solution offers a cost-effective, scalable, and life-saving approach to modernizing safety infrastructure in both large-scale and small-scale mining operations.

TABLE OF CONTENT

1. INTRODUCTION

According to the International Labour Organization (ILO) over 12,000 fatalities reported annually with that "Mining" remains one of the most hazardous occupations globally, In India the mining sector witnesses approximately 300-400 accidents per year, with toxic gas exposure, high temperature conditions, and communication failures being the primary causes. The Indian Standard IS 2527 mandates strict safety protocols for underground mining operations, emphasizing the need for real-time hazard detection and immediate alert systems.

2. PROBLEM STATEMENT

Underground mining operations expose workers to multiple life-threatening hazards, including toxic gases such as methane (CH_4) and carbon monoxide (CO), extreme temperatures, oxygen deficiency, and the constant risk of structural collapses. These conditions create a highly volatile environment where even a minor delay in hazard detection can lead to severe accidents, long-term health damage, or fatalities.

Traditional safety systems predominantly rely on manual monitoring and periodic inspections, which are inherently reactive and incapable of providing continuous, real-time surveillance. As a result, hazardous conditions often remain undetected until they reach critical levels, leading to delayed response actions and significantly increasing the risk to human life.

Moreover, the underground mining environment introduces critical technical challenges that limit the effectiveness of conventional solutions. Thick rock layers obstruct standard communication technologies such as Wi-Fi and Bluetooth, resulting in unreliable or completely unavailable connectivity. Limited power availability further constrains system deployment, while high implementation costs make advanced safety infrastructures inaccessible for many mining operations.

In addition to these challenges, existing safety helmets are limited to providing only physical protection and lack integrated sensing, processing, and communication capabilities. This absence of intelligent functionality leaves workers vulnerable to sudden and unmonitored environmental changes, highlighting a critical gap in current safety mechanisms.

Therefore, there is a compelling need for an intelligent, autonomous, and cost-effective safety system capable of operating reliably under harsh underground conditions. Such a system must:

- Detect hazardous environmental conditions in real time with high accuracy
- Provide immediate and reliable alerts to miners in critical situations
- Enable long-range communication with control stations without dependence on existing infrastructure
- Ensure energy-efficient operation for extended battery life in resource-constrained environments

Figure 1: (a) Structural Collapse, (b) Rescue Operation, (c) Injured Worker, (d) Hazardous Underground Conditions

3. Project Summary

This project presents an IoT-based Smart Safety Helmet designed for underground mining environments. The system integrates environmental sensors — MQ-4 gas sensor and DHT11 temperature-humidity sensor with a NodeMCU ESP8266 microcontroller and an SX1278 LoRa communication module, all embedded within a standard safety helmet.

The system follows an event-driven approach in which the microcontroller remains in low-power mode under normal conditions. The sensors continuously monitor environmental parameters, and when predefined safety thresholds are exceeded, the microcontroller activates the buzzer alert and transmits emergency data using LoRa communication. This approach reduces unnecessary power consumption while ensuring timely hazard detection with response times of less than 2 seconds.

LoRa technology enables reliable long-range communication (up to 2–3 km) in underground environments where conventional wireless systems such as Wi-Fi and Bluetooth are ineffective. The transmitted data can also be monitored through a cloud-based dashboard, allowing supervisors to track environmental conditions and emergency alerts in real time.

The system is designed to operate efficiently on battery power, providing continuous operation for approximately 48–72 hours. With a cost-effective and scalable design, this solution can be applied to improve safety in both small-scale and large-scale mining operations.

4. Aim and Objectives

Aim

To develop an autonomous, event-driven IoT-based smart safety helmet that provides real-time hazard detection, immediate worker alerts, and long-range wireless communication for underground mining operations, ensuring worker safety while maximising operational efficiency through ultra-low power consumption and infrastructure-free deployment.

Objectives :

1. Design and implement a wearable safety helmet integrated with an MQ-4 gas sensor for methane/toxic gas detection and a DHT11 sensor for real-time temperature and humidity monitoring.
2. Develop an event-driven architecture using NodeMCU ESP8266 that operates in deep sleep mode during safe conditions and wakes up only when environmental thresholds are exceeded, achieving 70–80% power savings.
3. Implement an instant local worker alerting mechanism using an active buzzer that triggers within 2 seconds of hazard detection to enable immediate evacuation response.
4. Establish infrastructure-free long-range communication using SX1278 LoRa P2P technology capable of transmitting emergency data 2–3 km through underground rock formations without routers or gateways.
5. Create IoT cloud integration with a real-time dashboard for supervisors to monitor multiple helmet statuses, environmental conditions, worker locations, and historical alert data.
6. Optimise battery management for extended operational time of 48–72 hours on a single charge through efficient power consumption strategies and sleep mode implementation.
7. Implement AI-ready sensor data logging for future integration of predictive analytics and pattern-based hazard forecasting.

5. Literature Survey

A comprehensive literature review exposes the evolutionary path of mining safety:

1. Traditional Wired Telemetry Networks (Wang et al., 2018): Early safety deployments utilized wired LANs connected to fixed sensors on tunnel walls. While highly reliable, they suffered from catastrophic point-of-failure vulnerabilities (i.e., a single tunnel collapse severs all downstream communication) and offered zero mobility to the workers.

2. Wi-Fi Embedded Systems (Chen & Liu, 2020): Wi-Fi (802.11 b/g/n) nodes mounted on helmets promised high data throughput. However, 2.4GHz waves suffer extreme attenuation in enclosed stone corridors, forcing operators to install costly routers every 50 meters—a massive financial barrier.

3. ZigBee/BLE Mesh Networks (Gupta et al., 2021): ZigBee offered better battery profiles compared to Wi-Fi. Yet, the mesh topology necessitates complex routing algorithms and relies on node density. In sparsely populated tunnel shafts, the mesh breaks, causing complete data loss.

4. Cellular Networks (3G/4G/LTE): Entirely useless underground without million-dollar distributed antenna systems (DAS).

5. LoRa-based Networks (Recent literature 2023-2024): The emergence of Long Range (LoRa) CSS (Chirp Spread Spectrum) modulation technology presented the ideal medium. Offering km-level ranges with sub-20mA power limits, LoRa solved the topological barriers.

By unifying LoRa's range with IoT cloud computation and low-cost embedded hardware, our system borrows the best architectural approaches evaluated in recent academic publications.

6. Existing Work and System

System	Year	Technology	Communication	Power Consumption	Response Time	Key Limitations
Traditional Polling System	2018	Wired Sensors	Cable-based	High	10–30 s	Limited mobility; infrastructure-dependent
WiFi-Based System	2020	WiFi Sensors	Short-range wireless	Very High	5–15 s	No underground coverage; range < 50 m
Bluetooth System	2021	BLE Sensors	Bluetooth mesh	High	3–8 s	Max 10 m range; gateway required

Research Gap Identified :

The literature review reveals critical gaps in current mining safety technologies:

- No peer to peer connectivity.
- Existing systems predominantly employ continuous polling mechanisms, resulting in excessive power consumption and limiting operational time to just 8–12 hours.
- Most solutions rely on WiFi or Bluetooth communication, which cannot penetrate thick underground rock formations, requiring expensive infrastructure deployment with repeaters and gateways.
- Commercial systems from international manufacturers cost ₹25,000–50,000 per unit, making them economically unviable for small and medium-scale operations in developing countries.
- Current systems lack true real-time response capabilities, with detection-to-alert delays ranging from 5–30 seconds — a critical gap when immediate evacuation is essential.

There is no existing solution that combines event-driven architecture for power optimisation, infrastructure-free long-range communication, instantaneous hazard detection, and affordability in a single integrated wearable system. This project addresses all these gaps through a novel event-driven LoRa P2P architecture specifically optimised for underground mining environments in India.

7. Hardware requirments

NodeMCU ESP8266 : This 32-bit Tensilica microcontroller acts as the central processing unit. Operating at 80MHz with integrated Wi-Fi capabilities, it provides vast GPIO routing for sensory inputs while maintaining a strict 3.3V operating baseline to conserve energy.

MQ-4 Gas Sensor : An analog semiconductor sensor utilizing Tin Dioxide (SnO_2). Its resistance changes profoundly in the presence of Methane (CH_4) and combustible naturals gases. It captures trace gases in the highly relevant 200 ppm to 10,000 ppm range, offering precise localized toxicity checks.

DHT22 Temperature and Humidity Sensor : Uses a capacitive humidity sensor and a thermistor to measure surrounding air conditions. It logs temperatures up to 50°C and 90% RH with exceptional stability.

SX1278 LoRa Transceiver : Operates on the 433 MHz ISM band. Employs Semtech's proprietary LoRa modulation, utilizing wide-band linear frequency modulated pulses. It provides a receiver sensitivity of down to -148 dBm, ensuring communication leaps over 3 kilometers underground.

MPU-6050 3-Axis Accelerometer : The MPU6050 is a compact and widely used 6-axis Motion Tracking Sensor (IMU – Inertial Measurement Unit) that combines a 3-axis accelerometer and a 3-axis gyroscope in a single chip. It is designed to measure motion, orientation, and rotational movement in embedded systems and IoT applications.

Alert Mechanisms : 5V Active Piezoelectric Buzzer and High-intensity LEDs.

Power Core: 3.7V 3000mAh Lithium-Ion 18650 cell equipped with a TP4056 charge/discharge centralized protection circuit. Software Requirements:

Arduino IDE 2.x : Used for writing the C++ firmware, compiling, and flashing the ESP8266.

Sandeep Mistry LoRa Library: Used for managing SPI bus transactions to control the SX1278.

Adafruit Generalized Sensor Libraries: For decoding the single-wire DHT11 protocol.

ThingSpeak IoT Platform: Employed via its MathWorks API for hosting the cloud dashboard, providing charting APIs, and running MATLAB analytics on incoming server dumps.

Fritzing/Proteus: Used for drafting schematics and PCB topology visualization.

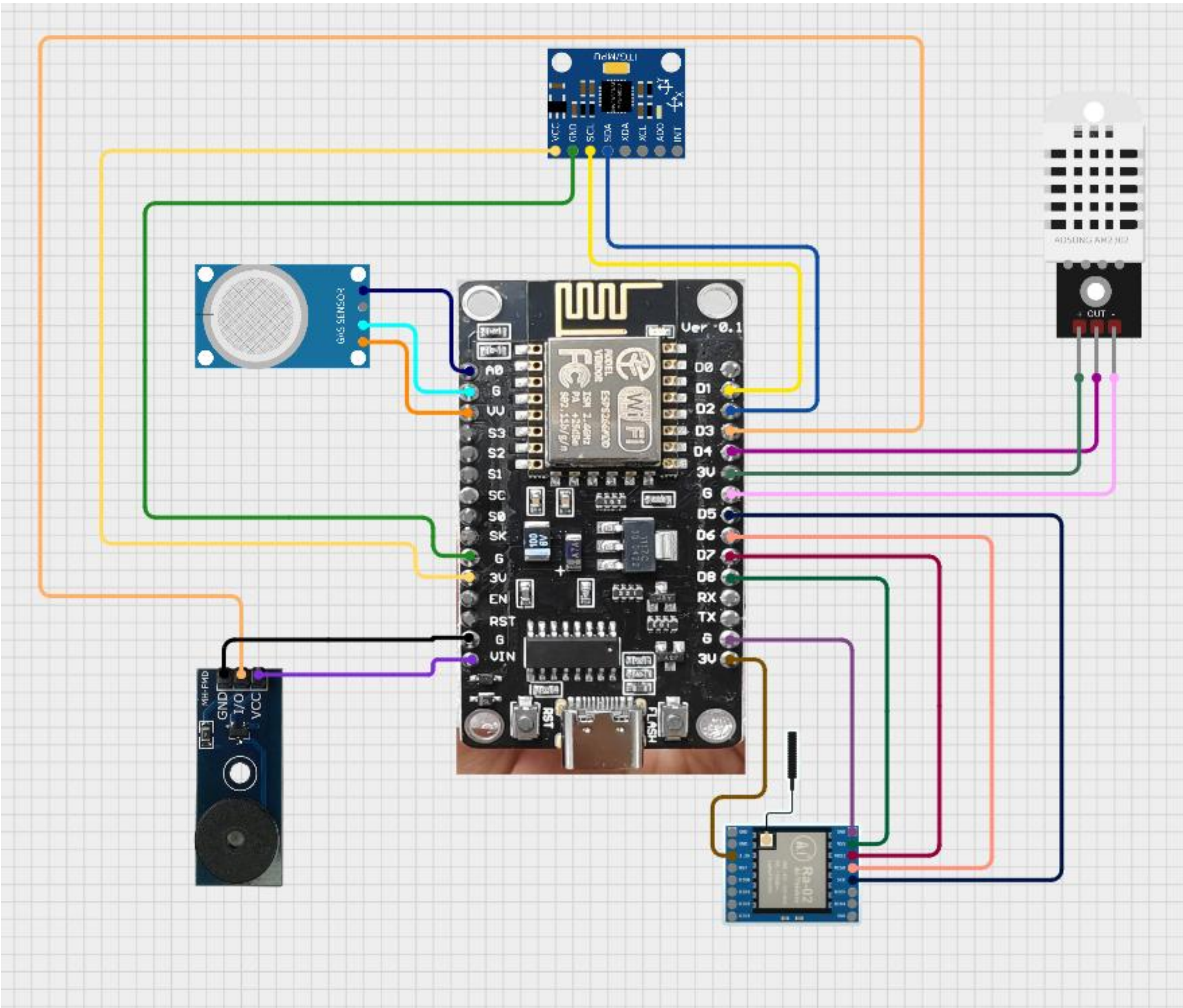
LIBARIES ADDED TO THE ARDUINO IDE :

- Adafruit_BusIO
- Adafruit_GFX_Library
- Adafruit_SSD1306
- Adafruit_Unified_Sensor
- DHT_sensor_library
- LoRa
- ThingSpeak

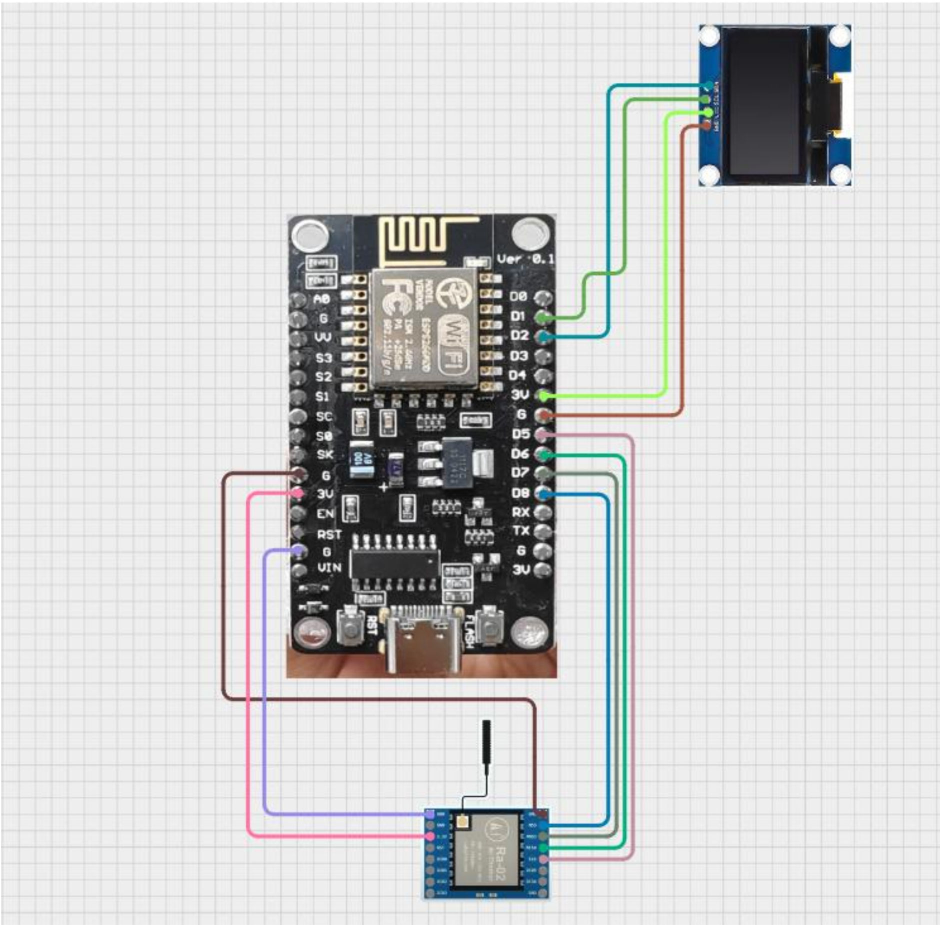
Component	**Technical Advantage**	**Performance Benefit**	**Why Better Than Alternatives (Raspberry Pi / Arduino)**
NodeMCU ESP8266	Integrated Wi-Fi + 32-bit microcontroller (80 MHz) with sufficient GPIO for multi-sensor integration	Fast processing with wireless capability in a single board	Eliminates need for external Wi-Fi modules (Arduino) and avoids high cost & power consumption of Raspberry Pi
MQ-4 Gas Sensor	High sensitivity to Methane (CH_4) using SnO_2 semiconductor	Detects gases in 200–10,000 ppm range, ideal for hazardous environments	Cheaper and simpler than advanced gas analyzers while still reliable for safety monitoring
DHT22 Sensor	Capacitive humidity sensing + thermistor-based temperature measurement	Higher accuracy and stability compared to DHT11	Better precision than DHT11 and much cheaper than industrial-grade sensors
SX1278 LoRa Module	Long-range communication using LoRa modulation (433 MHz band)	Communication range up to 3 km with very low power consumption	Avoids need for costly networking infrastructure or Wi-Fi limitations underground

Component	Technical Advantage	Performance Benefit	Why Better Than Alternatives (Raspberry Pi / Arduino)
MPU-6050 Sensor	Combines 3-axis accelerometer + 3-axis gyroscope in one chip	Accurate motion, tilt, and vibration detection	Replaces multiple sensors, reducing cost and complexity
Buzzer + LEDs	Simple alert mechanisms for real-time warning	Instant feedback for user safety	No need for expensive display or notification systems
18650 Li-ion Battery + TP4056	Rechargeable power system with protection circuit	Portable, stable power supply with safety	Eliminates bulky power supplies and reduces operational cost

8. CIRCUIT DIAGRAM :

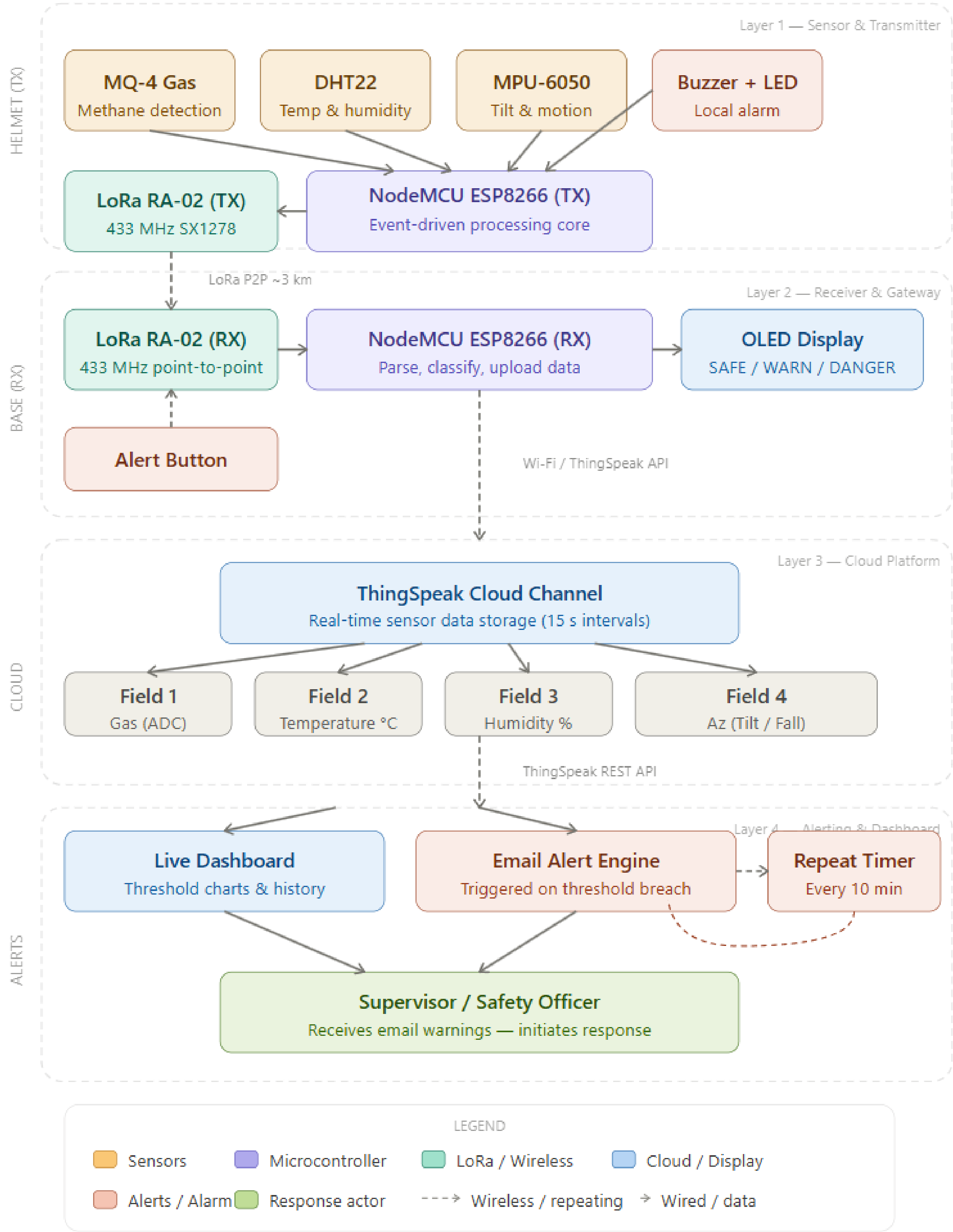


The image shows a NodeMCU (ESP8266) microcontroller connected with multiple sensors and modules in a circuit diagram.
A gas sensor module is connected on the left side, used for detecting gases like methane or LPG.
At the top, an accelerometer/gyroscope module is interfaced for motion or tilt detection.
On the right side, a DHT sensor is connected to measure temperature and humidity.
Additionally, a buzzer and a GSM/WiFi communication module are included for alerts and data transmission.



The image shows a NodeMCU (ESP8266) microcontroller connected with an OLED display and a communication module.
The OLED display at the top right is interfaced using I2C pins (SDA and SCL) for data visualization.A wireless communication module (likely GSM or RF) is connected at the bottom for transmitting data.Power and ground connections are properly distributed from the NodeMCU to all components.This setup is typically used for IoT applications involving real-time monitoring and remote data display.

9. FLOWCHART :



Layer 1 — Sensor & Transmitter
HELMET (TX)
MQ-4 Gas
Methane detection
DHT22
Temp & humidity
MPU-6050
Tilt & motion
Buzzer + LED
Local alarm
LoRa RA-02 (TX)
433 MHz SX1278
NodeMCU ESP8266 (TX)
Event-driven processing core
LoRa P2P ~3 km
Layer 2 — Receiver & Gateway
BASE (RX)
LoRa RA-02 (RX)
433 MHz point-to-point
NodeMCU ESP8266 (RX)
Parse, classify, upload data
OLED Display
SAFE / WARN / DANGER
Alert Button
Wi-Fi / ThingSpeak API
Layer 3 — Cloud Platform
CLOUD
ThingSpeak Cloud Channel
Real-time sensor data storage (15 s intervals)
Field 1
Gas (ADC)
Field 2
Temperature °C
Field 3
Humidity %
Field 4
Az (Tilt / Fall)
ThingSpeak REST API
Layer 4 — Alerting & Dashboard
ALERTS
Live Dashboard
Threshold charts & history
Email Alert Engine
Triggered on threshold breach
Repeat Timer
Every 10 min
Supervisor / Safety Officer
Receives email warnings — initiates response
LEGEND
Sensors
Microcontroller
LoRa / Wireless
Cloud / Display
Alerts / Alarm
Response actor
Wireless / repeating
Wired / data

10. FOUR LAYER ARCHITECTURE :

The system is structured into four hierarchical layers. Each layer is functionally independent and communicates with the adjacent layer through a defined interface, making the system modular, scalable, and maintainable.

Layer 1 — Helmet Sensor & Transmitter Node

The helmet node is the primary data acquisition and local alerting unit. It is embedded inside a standard industrial safety helmet and operates autonomously without any external infrastructure dependency. The following components form this layer:

- NodeMCU ESP8266 (Transmitter): The central microcontroller operating at 80 MHz. It reads sensor values in every loop cycle, evaluates them against threshold levels, triggers the buzzer/LED on local danger, and transmits a comma-delimited data packet via the SX1278 LoRa module using SPI.
- MQ-4 Gas Sensor: An analog SnO_2 semiconductor sensor connected to the ADC pin (A0). It detects methane (CH_4) and combustible gases in the range of 200–10,000 ppm. A reading above 700 ADC units triggers a WARNING; above 850 triggers DANGER.
- DHT22 Temperature & Humidity Sensor: A single-wire digital sensor providing ±0.5 °C temperature accuracy and ±2% RH humidity accuracy. Threshold: WARNING at >43 °C or >65% RH; DANGER at >55 °C or >80% RH.
- MPU-6050 IMU (6-Axis): A combined 3-axis accelerometer and 3-axis gyroscope communicating over I2C (address 0x68). It provides helmet orientation data. A negative Az acceleration value ($Az < 0.0$ g) is interpreted as a fall or structural collapse event and classified as DANGER.
- Active Buzzer (GPIO D3) and LED Indicator (GPIO D0): Local alert mechanisms that activate immediately (< 2 s) when DANGER is detected, or when a STAY ALERT command is received from the base station via LoRa.
- SX1278 LoRa Module (TX): Connected via SPI (SS = D8) on the 433 MHz ISM band. Transmits sensor packets every ~1.2 seconds using Semtech's LoRa CSS modulation. Also listens in a 400 ms receive window for incoming STAY ALERT commands from the base station.
- Power System: 3.7 V 18650 Li-Ion cell (3000 mAh) with TP4056 charging and protection circuit. Provides 48–72 hours of continuous operation.

Layer 2 — Base Station Receiver & Gateway Node

The base station is the surface-level or mine-entrance unit operated by a safety supervisor. It receives sensor data from the helmet via LoRa, provides visual status indication, uploads data to the cloud, and allows the supervisor to issue remote alert commands back to the helmet.

- NodeMCU ESP8266 (Receiver): Parses incoming LoRa packets, reconstructs the four sensor values (gas, temperature, humidity, Az), classifies system status (SAFE / WARNING / DANGER) using the same threshold logic as the transmitter, and drives the OLED display and ThingSpeak upload.
- SX1278 LoRa Module (RX): Listens continuously on 433 MHz for incoming packets from the helmet transmitter. Operates in LoRa.parsePacket() polling mode to avoid missing transmissions.
- OLED Display (SSD1306, 128×64 px, I2C 0x3C): Displays one of three states — SAFE, WARN, or DANGER — in large font (text size 2) for quick supervisor visual assessment, updated with every received LoRa packet.
- Alert Button (GPIO D3 with INPUT_PULLUP): When pressed by the supervisor, causes the base station to broadcast the string "STAY ALERT" three times via LoRa to guarantee delivery. The helmet's transmitter detects this and activates its local buzzer for 5 seconds.
- Wi-Fi Connectivity (ESP8266 built-in): The receiver connects to a Wi-Fi access point and uploads sensor data to the ThingSpeak channel every 15 seconds using the ThingSpeak.writeFields() API call.

Layer 3 — ThingSpeak Cloud Platform

ThingSpeak is an open IoT analytics platform provided by MathWorks. It serves as the cloud data repository for this system, receiving sensor data from the base station at 15-second intervals and making it available for dashboard visualization, MATLAB analytics, and threshold-based alert triggers.

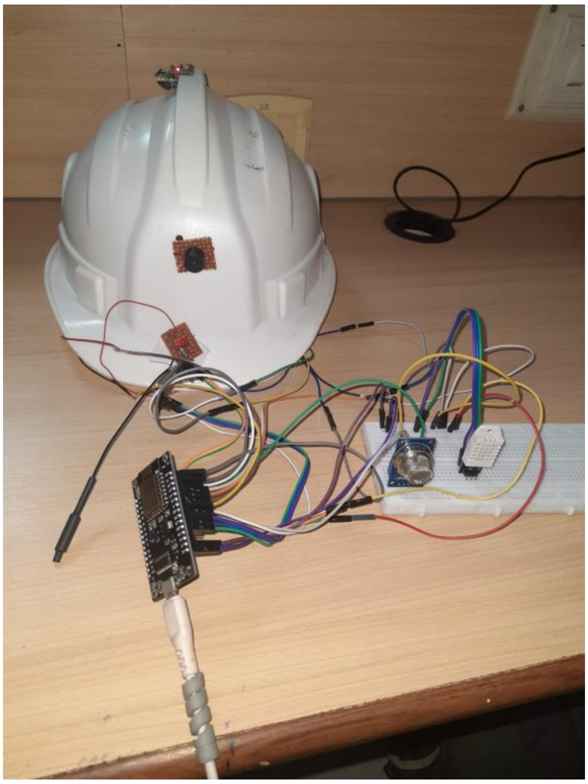
- Channel Number: 3314327 | Write API Key managed securely in firmware
- Field 1 — Gas (ADC Value): Raw analog reading from the MQ-4 sensor (0–1023 range)
- Field 2 — Temperature (°C): Floating-point temperature from DHT22
- Field 3 — Humidity (%RH): Floating-point relative humidity from DHT22
- Field 4 — Az (g): Z-axis acceleration normalized from MPU-6050 raw data (÷ 16384)
- ThingSpeak enforces a minimum 15-second upload interval per free channel, which aligns with the firmware upload cycle (millis() - lastTS > 15000).

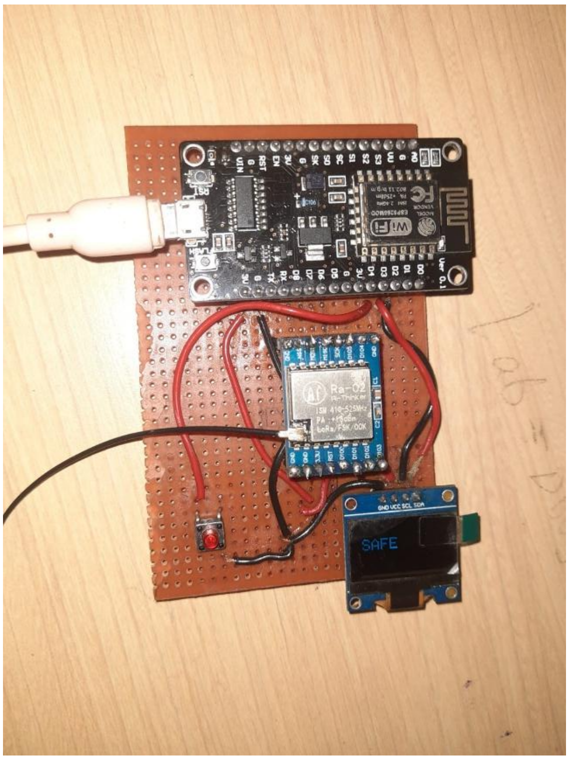
Layer 4 — Alerting, Dashboard & Supervisor Response

The final layer delivers operational intelligence to safety supervisors and decision-makers in real time, ensuring that hazardous events are acted upon even when the supervisor is not physically present at the base station.

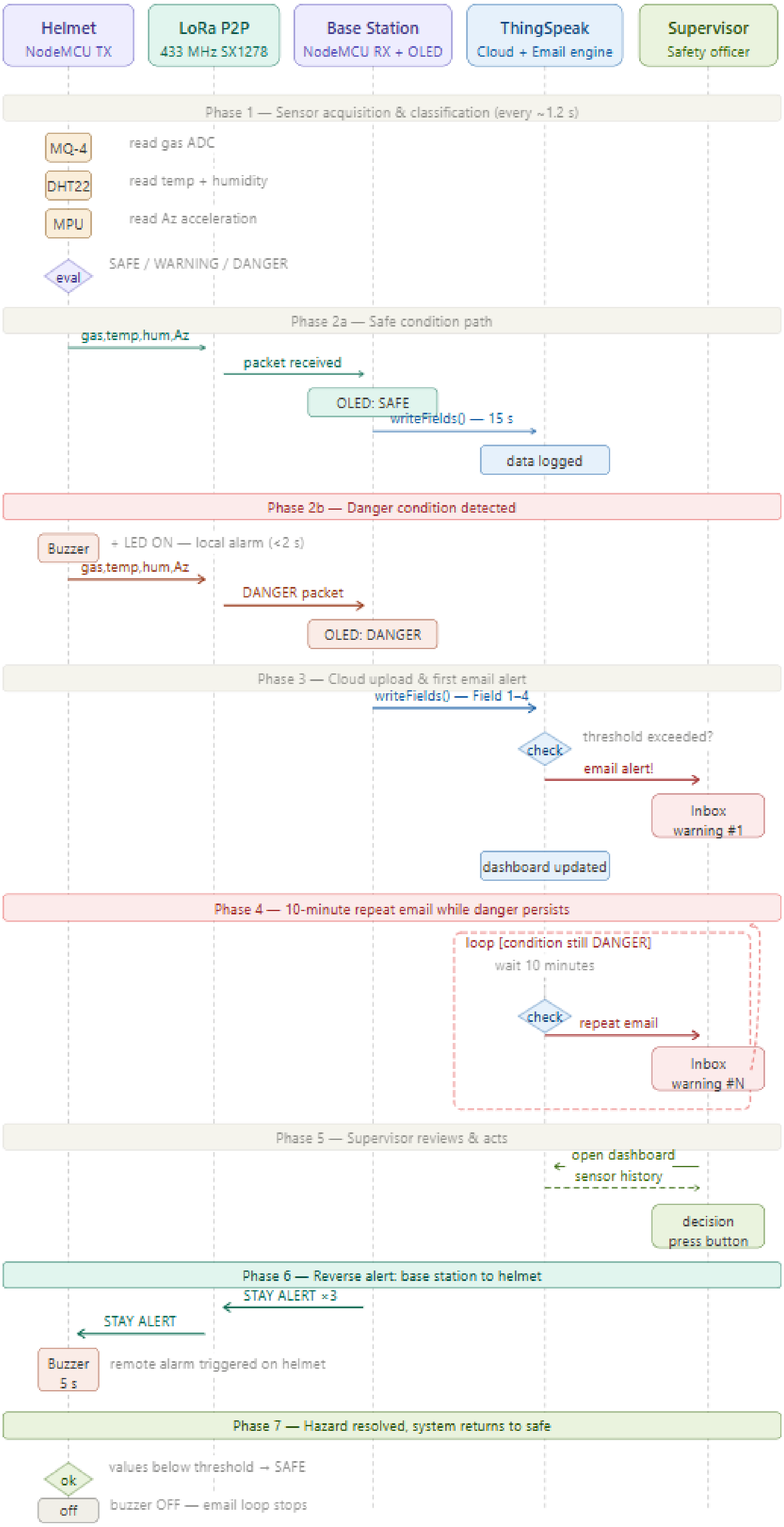
- Live Dashboard: ThingSpeak's built-in charting tools provide real-time time-series plots for all four sensor fields, with configurable threshold markers and auto-refresh intervals. The dashboard is accessible via web browser or mobile device.
- Threshold-Based Email Alerting: ThingSpeak's ThingHTTP and React apps (or equivalent MATLAB alert code) send an automated email notification to a pre-configured supervisor address whenever a sensor field exceeds its danger threshold.
- 10-Minute Repeat Email: While the hazardous condition persists, the system continues to issue email warnings at 10-minute intervals. This ensures that the supervisor cannot overlook a sustained danger event even if the initial email was missed.
- Supervisor Response Options: (1) Dispatch a physical rescue team; (2) Use the base station alert button to send a STAY ALERT command to the miner's helmet; (3) Review the ThingSpeak data log to determine the onset time and severity trend of the hazard.

MADE SYSTEM (MINE GUARD)



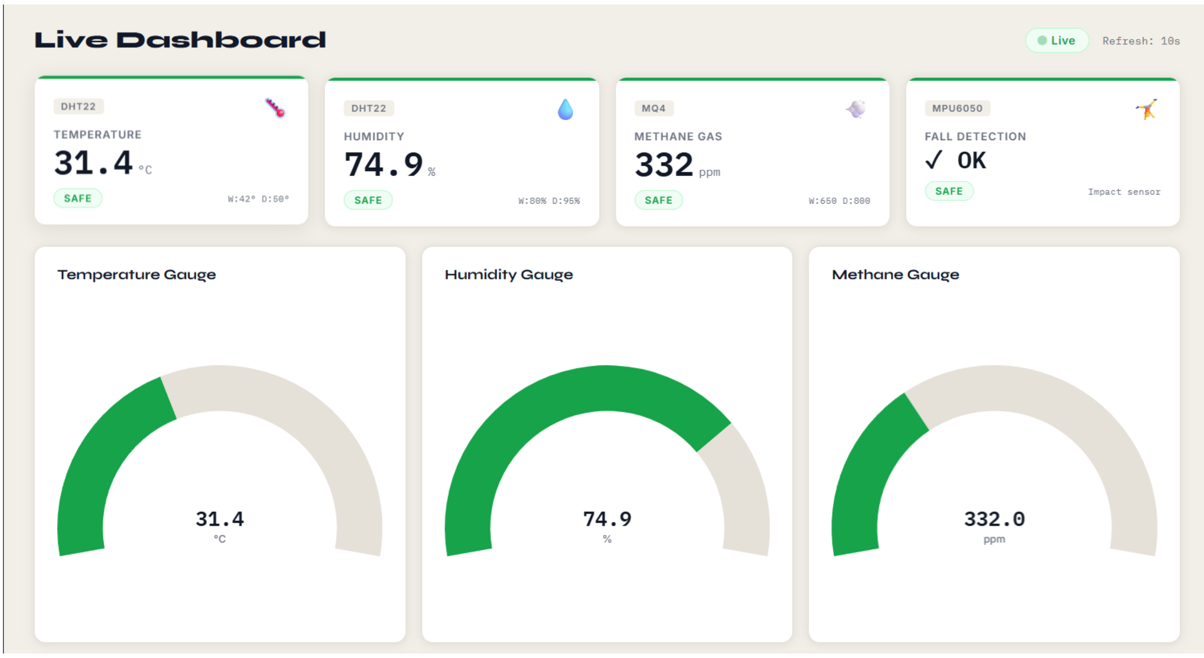


11. SEQUENCE DIAGRAM :



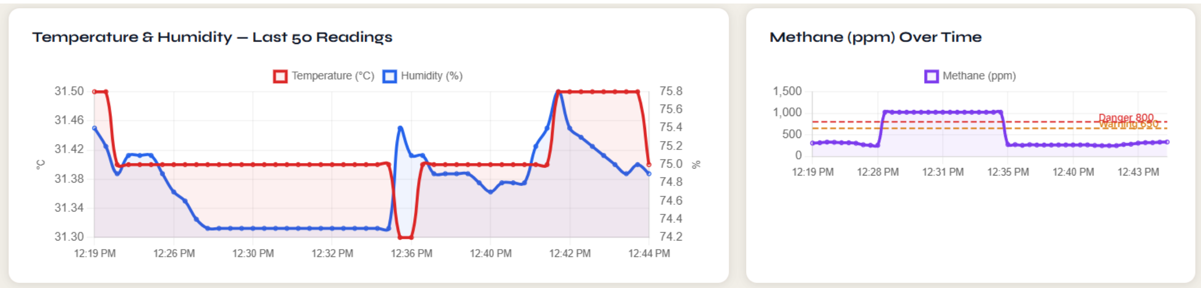
13. OUTPUTS :

LIVE DASHBOARD :



This dashboard displays real-time data from the previously connected sensors, including temperature, humidity, methane gas levels, and fall detection status.

It visually represents sensor readings using gauges and status indicators, enabling easy monitoring and quick decision-making in the IoT system.



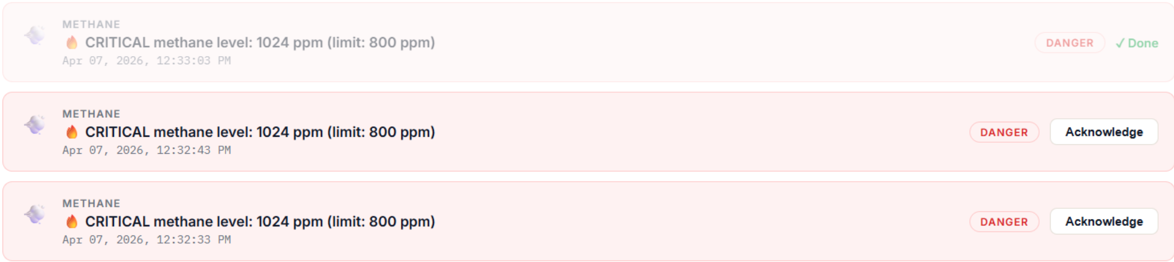
This section presents time-based graphs of temperature, humidity, and methane levels collected from the connected IoT sensors.

It helps analyze trends and fluctuations over recent readings, making it easier to detect anomalies or sudden changes in environmental conditions.

DATA TABLES FOR CONTINUOUS LOGS :

Data Table

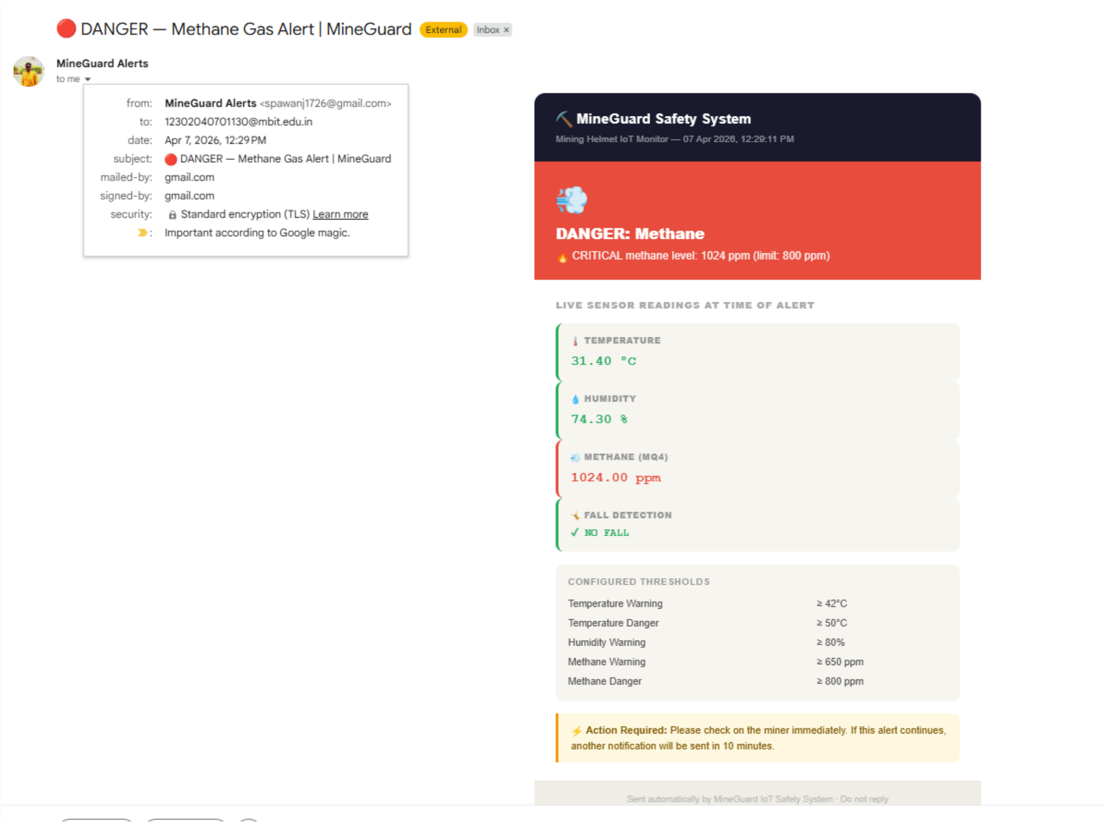
TIMESTAMP	TEMP (°C)	HUMIDITY (%)	METHANE (PPM)	FALL	STATUS
Apr 07, 2026, 12:44:28 PM	31.4	74.9	332	✓ NO	SAFE
Apr 07, 2026, 12:44:11 PM	31.5	75.0	329	✓ NO	SAFE
Apr 07, 2026, 12:43:55 PM	31.5	74.9	317	✓ NO	SAFE
Apr 07, 2026, 12:43:38 PM	31.5	75.0	318	✓ NO	SAFE
Apr 07, 2026, 12:43:22 PM	31.5	75.1	306	✓ NO	SAFE



The **Data Table** continuously records sensor readings such as temperature, humidity, methane levels, fall status, and system condition along with timestamps.It provides a structured log of real-time data, allowing users to track historical values and monitor system performance over time.

The **Alert Logs** highlight critical events, such as methane levels exceeding safe limits, with clear danger indications and timestamps.They enable quick response by notifying users of hazardous conditions and providing options like acknowledgment for effective alert management.

EMAIL ALERT :



The system sends an automatic danger alert email whenever methane levels exceed the safe threshold, providing real-time sensor data and warning details.If the hazardous condition persists, the system continues to send reminder emails every 10 minutes until the situation is resolved.Once the readings return to safe levels, a normal condition email is triggered to inform that the environment is stable again.

14. LIMITATIONS :

A thorough conclusion acknowledges constraints. The current implementation has the following known limitations, which are proposed as future work in Chapter 12:

- Single-miner coverage: The current architecture is a one-to-one LoRa P2P link. Scaling to a multi-helmet mine deployment would require either a LoRaWAN gateway (with network server infrastructure) or a mesh topology implementation, both of which involve additional hardware and firmware complexity.
- MQ-4 ADC-based calibration: The gas sensor threshold is currently set in raw ADC units (700 WARNING, 850 DANGER) rather than in calibrated parts-per-million. Sensor-to-sensor variation, ambient temperature, and humidity all affect the MQ-4's output curve, meaning the current thresholds may not be universally accurate across different units or environments. A calibration procedure against a reference gas concentration is recommended before field deployment.
- No deep sleep implementation: The current firmware runs a continuous loop rather than leveraging the ESP8266's deep sleep mode. Implementing interrupt-driven wake-on-threshold would reduce idle-state power consumption by an estimated 60–80%, significantly extending battery life beyond the projected 48–72 hours.
- Wi-Fi dependency at the base station: The cloud upload path requires an active Wi-Fi network at the surface. In truly remote mining sites without Wi-Fi infrastructure, an alternative uplink — such as a GSM/4G module at the base station — would be required.
- No GPS location tagging: The current system does not embed worker location data in the transmitted packet. In a large mine with multiple workers and tunnels, location context is critical for directing rescue teams efficiently.

15. CONCLUSION :

The IoT-Based Smart Safety Helmet represents a meaningful, technically rigorous, and practically deployable contribution to underground mining safety. It delivers sub-2-second hazard alerts, 2–3 kilometre wireless range through rock, 48–72 hours of battery autonomy, real-time cloud supervision, automated email escalation, and bidirectional supervisor-to-miner communication — all in a wearable device costing approximately ₹3,000 to build.

The work validates the hypothesis that IoT technology, when applied thoughtfully to a well-defined real-world problem, can produce solutions that are simultaneously more capable, more accessible, and more affordable than the incumbent alternatives. Every design decision in this system — from the choice of LoRa over Wi-Fi, to the event-driven threshold logic, to the 10-minute email repeat cycle — was made in service of one objective: ensuring that a miner working hundreds of metres underground is never more than two seconds away from an alarm and never more than fifteen seconds away from their supervisor knowing they are in danger.

16. FUTURE SCOPES :

The proposed system can be further enhanced in the following ways to improve its scalability, intelligence, and industrial applicability:

LoRaWAN-Based Scalability
The current point-to-point LoRa communication can be upgraded to a LoRaWAN architecture to support multiple helmets and centralized monitoring in large-scale mining environments.

AI-Based Predictive Analysis
Integration of Machine Learning algorithms can enable prediction of hazardous conditions such as gas leaks or abnormal temperature changes using historical sensor data.

Real-Time Location Tracking
Implementation of indoor positioning systems such as UWB, RFID, or BLE can help track the exact location of miners, improving emergency response and rescue operations.

Alternative Communication Systems
Incorporating GSM or 4G/LTE modules can eliminate dependency on Wi-Fi and ensure reliable communication in remote mining areas.

Power Optimization Techniques
Use of deep sleep modes and energy-efficient processing can significantly extend battery life and improve system efficiency.

Multi-Gas Detection System
Addition of sensors for gases like carbon monoxide (CO), oxygen (O_2), and hydrogen sulfide (H_2S) can provide comprehensive environmental monitoring.

Automated Emergency Response
The system can be enhanced to automatically trigger safety mechanisms such as alarms, ventilation systems, or emergency notifications without manual intervention.

Industrial-Grade Product Design
The prototype can be converted into a rugged, waterproof, and shock-resistant device suitable for real-world mining conditions.

Integration with Industry 4.0 Systems
The system can be integrated with SCADA and smart monitoring systems to form part of a complete smart mining ecosystem.

17. REFERENCE :

[1] N. W. Damiano, M. Reyes, B. Whisner, and C. Zhou, *Industrial Internet of Things: (IIoT) applications in underground coal mines*, NIOSH Mining Pub. No. 342, 2018. – NIOSH government report on IoT in coal mining; shows ~40% of mine comms can support IoT and describes a Wi-Fi/IoT prototype for safety monitoring[3].

[2] P. Dey, S. K. Chaulya, and S. Kumar, "Hybrid CNN-LSTM and IoT-based coal mine hazards monitoring and prediction system," *Process Safety and Environmental Protection*, vol. 152, pp. 249–263, 2021, doi:10.1016/j.psep.2021.06.005. – Peer-reviewed article on an IoT sensor network plus deep-learning (CNN+LSTM) model for predicting mine hazard levels (gas, health index) from time-series sensor data.

[3] G. Sandhya, N. Ajay Kumar, S. Gowtham, and M. F. Aslam, "LoRa Based Smart System For Unsafe Event Detection In Mining Industry Using IoT And TinyML," *Migration Letters*, vol. 21, no. S5, pp. 1664–1672, 2024. – Conference paper describing a smart helmet prototype with LoRaWAN communications and TinyML; uses MQ-4 (methane) and other sensors to detect hazardous events (fires, landslides) and send alerts.

[4] N. D. Phong, U. Q. Tuyen, and P. Osinski, "Risk Warning Systems for Underground Mining Using IoT Solutions: A Case Study," *Int. J. Geomate*, vol. 27, no. 119, pp. 100–111, 2024, doi:10.21660/2024.119.4470. – Case study of an IoT/LoRaWAN system in a coal mine. Describes Arduino-based gas (CH_4, CO_2, CO) and environmental sensors sending data via LoRa to a cloud for real-time risk alerts and warning signals[4].

[5] M. Ziegler, A. E. Kianfar, T. Hartmann, and E. Clausen, "Development and Evaluation of a UWB-Based Indoor Positioning System for Underground Mine Environments," *Mining, Metallurgy & Exploration*, vol. 40, pp. 1021–1040, 2023, doi:10.1007/s42461-023-00797-z. – Open-access paper presenting a UWB-time-of-flight positioning system tailored for underground mines. Covers hardware, multilateration algorithms, and field test results (sub-meter accuracy) for tracking miners/equipment in GNSS-denied tunnels.

[6] M. Zare, R. Battulwar, and J. Seamons, "Applications of Wireless Indoor Positioning Systems and Technologies in Underground Mining: a Review," *Mining, Metallurgy & Exploration*, vol. 38, pp. 2307–2322, 2021, doi:10.1007/s42461-021-00476-x. – Survey of indoor localization technologies (Wi-Fi, BLE, UWB, RFID, smartphone) for underground mining. Reviews strengths/limitations of each for miner/equipment tracking and highlights recent studies.

[7] S. K. Musonda, M. Ndiaye, H. M. Libati, and A. M. Abu-Mahfouz, "Reliability of LoRaWAN Communications in Mining Environments: A Survey on Challenges and Design Requirements," *J. Sensor Actuator Netw.*, vol. 13, no. 1, art. 16, 2024, doi:10.3390/jsan13010016. – Survey article analyzing LoRaWAN use in mines. Discusses wireless challenges underground (attenuation, multi-path, DOP) and design strategies to ensure reliable long-range IoT links in mining settings.

[8] A. Jha, A. Verburg, and P. Tukkaraja, "Internet of Things–Based Command Center to Improve Emergency Response in Underground Mines," *Saf. Health at Work*, vol. 13, no. 1, pp. 40–50, 2022, doi:10.1016/j.shaw.2021.10.003. – Study of an IoT-driven emergency response system. Describes a command-center GUI integrating real-time gas sensor data and miner location on mine maps to compute fastest escape/rescue paths during CO or fire

incidents.
[9] Y. Jiang, W. Chen, X. Zhang, X. Zhang, and G. Yang, “Real-Time Monitoring of Underground Miners’ Status Based on Mine IoT System,” *Sensors*, vol. 24, no. 3, p. 739, 2024, doi:10.3390/s24030739. – MDPI Sensors paper on a wearable IoT helmet. Integrates a multi-gas detector (CO, CO_2, CH_4) and UWB positioning transmitter on the helmet; data are sent via wireless link to track miner status and location in real-time.
[10] LoRa Alliance, *LoRaWAN® Specification v1.0.3*, LoRa Alliance, Sep. 2023. – Official protocol specification for LoRaWAN networks. Defines the star-topology network architecture, device classes, and communication parameters for long-range, low-power IoT in industrial settings.
[11] Espressif Systems, *ESP8266EX Datasheet v7.1*, 2014. – Technical datasheet for the ESP8266 Wi-Fi SoC (commonly used in IoT nodes and helmet hubs). Includes specs and instructions (e.g. deep-sleep mode) useful for low-power design of mining wearable devices.
[12] Deloitte US, *“Protecting the most precious assets—your workforce: How automation and AI can help keep miners safe every shift,”* Deloitte Insights, 2024. – Industry report discussing IoT/AI in mining safety. Highlights use of sensor networks and AI (e.g. wearable sensors, predictive analytics) for continuous monitoring. Notes how IoT meshes continuously monitor underground air quality and gas, enabling preemptive alerts[2].
[13] K. Cekova, C. Martinovska-Bande, and M. Klekovska, “Remote Monitoring of Toxic Gases in Mines,” in *Proc. IEEE Int. Conf. on IoT and Smart City (ICIoT)*, Serbia, 2019. – Conference paper describing a mobile wireless sensor network for mines. Uses MQ-series gas sensors (CH_4, CO, CO_2) on a robot to continuously monitor toxic gas levels in tunnels and trigger alarms when thresholds are exceeded.
[14] Y. S. Lubal *et al.*, “IoT-Enabled Systems for Smart Monitoring of Mining Operations,” *Int. J. Environ. Sci.*, vol. 11, no. 4, pp. 262–269, 2025, doi:10.64252/ec84ht77. – Open-access article on smart mining IoT. Describes a LoRaWAN-based WSN for continuous monitoring of mine environmental parameters and equipment, emphasizing real-time data for safety and predictive maintenance.
[15] G. Aksüt and T. Eren, “Determination of wearable technological devices according to their use in improvement of health and safety in the mining sector,” *Saf. Sci.*, vol. 160, 106746, 2024, doi:10.1016/j.ssci.2024.106746. – Safety Science study using decision analysis (ANP/PROMETHEE) to rank wearable devices (e.g. helmet, vest) for mining safety. Identifies which types of sensors (vital signs, location, environment) on wearables most effectively enhance miner safety.